

METHODOLOGY FLOWCHART • 11-STATE TOMATO GREENHOUSE CONTROL SYSTEM

Complete Scientific Pipeline: Plant Biology & Greenhouse Anatomy → Soil Physics → Complete Equilibrium Trim → Linearization → Controllability & Observability → Tri-Hybrid Synthesis → 24h Benchmark

Crop: *Solanum lycopersicum*

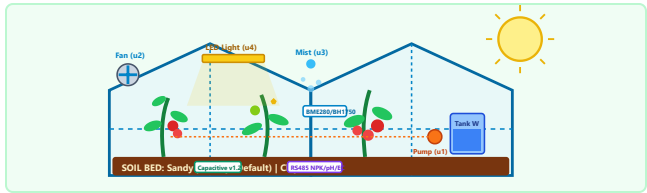
100% Pure Base MATLAB

Zero External Toolboxes

Master Kernel: main.m

1. GREENHOUSE ANATOMY & SENSORS

ESP32 Transducer Suite



Mathematical State Vector ($x \in \mathbb{R}^{11}$):

$[M, T, RH, L, pH, EC, N, P, K, W, VOC]^T \cdot 4 \text{ Controls } u \in [0, 1]^4$

3. COUPLED PLANT EQUATIONS

$dx/dt = f(x, u, d)$

Climate State Derivatives:

$$dM/dt = \gamma \cdot u_1 - E_{\text{loss}} - D$$

$$D = k_{\text{drain}} \max(M - M_{\text{th}}, 0)$$

$$E_{\text{loss}} = k_{\text{evap}} E [1 + 0.01 \max(T - 25, 0)]$$

$$dT/dt = 0.15(T_{\text{ext}} - T) + 0.00008 L - 3.0 u_2$$

$$dRH/dt = 8.0 u_3 - 4.0 u_2 - 0.15 \max(T - 25, 0)$$

$$dL/dt = 0.5(L_{\text{nat}} - L) + 30,000 u_4$$

Soil Substrates:

• **Sandy:** $k_d=0.045$, $\gamma=14.0$, $k_{\text{evap}}=2.60$

• **Loamy:** $k_d=0.020$, $\gamma=12.0$, $k_{\text{evap}}=2.00$

• **Clay:** $k_d=0.008$, $\gamma=9.5$, $k_{\text{evap}}=1.40$

Physical Meaning: Non-linear couplings: ventilation simultaneously cools (u_2) and dehumidifies ($-4u_2$); lighting (u_4) adds heat (+0.00008L).

5. COMPLETE EQUILIBRIUM TRIM

Gate: $\|dx_0\| \approx 0$

A. Environmental Inversion ($dx_{1:4}/dt = 0$):

$$u_1^* = [E_{\text{loss}} + D] / \gamma_{\text{irr}} = 15.0\% \text{ (Pump)}$$

$$u_2^* = [0.15(T_{\text{ext}} - T^*) + 0.00008L^*] / 3 = 40.0\% \text{ (Fan)}$$

$$u_3^* = [4 u_2^* - (-0.15 \max(T^* - 25, 0))] / 8 = 20.0\%$$

$$u_4^* = \max(0, L^* - L_{\text{nat}}) / 30000 = 0.0\% \text{ (Light)}$$

B. Chemical Stationary Equilibrium:

$$N^* = 132.8 \text{ mg/kg}, \quad P^* = 48.2 \text{ mg/kg}, \quad K^* = 189.5 \text{ mg/kg}$$

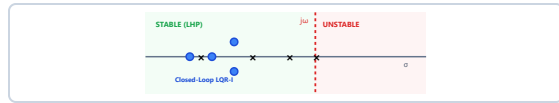
$$pH^* = 6.37, \quad EC^* = 2.41 \text{ dS/m}, \quad W^* = 80\%, \quad VOC^* = 100$$

VERIFICATION GATE: $\|dx_0/dt\|_{\infty} = 1.25 \times 10^{-10}$

Proof: Exact equilibrium across all 11 states simultaneously. Eliminates non-equilibrium linearization drift!

7. MODAL MAP & CONTROLLABILITY

Kalman Decomposition



Controllability Matrix:

$$Co = [B, AB, \dots, A^{10}B] \Rightarrow \text{rank}(Co) = 4 / 11$$

Proof: 4 climate states directly controllable. 7 substrate modes are passively stabilizable via self-dissipation.

9. TRI-HYBRID CONTROL LAWS

PID • Pole Placement • LQR-I

1. Decentralized PID + Anti-Windup Clamping:

$$u_{\text{raw}} = K_p e + K_i \int e \, dt + K_d \frac{de}{dt}$$

Anti-Windup: Freeze $\int e \, dt$ if $(u \geq 1 \wedge e > 0) \vee (u \leq 0 \wedge e < 0)$.

2. Scaled Subspace Pole Placement:

$$(A_{\text{env}} - \lambda_j I) x_j = B_{\text{env}} g_j \Rightarrow K_{\text{env}} = G X^{-1}$$

Bound: $|K_{ij}| \leq 0.2167$ (Erases 10^7 gain explosion!).

3. Optimal LQR-I via Real Schur CARE:

$$H = [A_a, -B_a R^{-1} B_a^T; -Q_a, -A_a^T] \in \mathbb{R}^{30 \times 30}$$

$$[U, T] = \text{schur}(H); \text{ordschur}(U, T, 'lhp'); P = \text{Re}(V_2 V_1^{-1})$$

$$u = u_0^* - K_x \delta y - K_i z_{\text{int}} \bullet \text{Res} = 1.25 \times 10^{-10}$$

Asymptotic Zero Error: Integrator state z_{int} eliminates disturbance offsets across all 5 growth phases.

2. BOTANICAL GROWTH PHASES

FAO-56 & UC Davis VRIC



Growth Stage	T _{opt} (°C)	RH _{opt} (%)	M _{opt} (%)	L _{opt} (lux)
1. Germination (0-2h)	24.0	72.5	70.0	10,000
2. Vegetative (2-8h)	25.0	65.0	60.0	18,000
3. Flowering (8-12h)	23.5	62.5	58.0	22,500
4. Fruit Dev. (12-18h)	25.0	60.0	65.0	22,500
5. Maturity (18-24h)	22.5	55.0	55.0	20,000

4. TRANSPIRATION & NUTRIENTS

NPK Kinetics

Transpiration Crop Demand:

$$\tau = [0.50 + 0.50(L/25000)] \times [1 + 0.02 \max(T - 22, 0)]$$

Macronutrient Fertiligation & Leaching:

$$dN/dt = 22 u_1 - 3.8\tau - 0.05 L_{\text{fac}} D (N/100) + 0.03(130 - N)$$

$$dP/dt = 7.5 u_1 - 1.2\tau - 0.02 L_{\text{fac}} D (P/50) + 0.02(48 - P)$$

$$dK/dt = 28 u_1 - 4.8\tau - 0.04 L_{\text{fac}} D (K/150) + 0.03(190 - K)$$

Tank & VOC Balance:

$$dW/dt = [0.05(80 - W) + q_{\text{refill}}] - (0.5E + 0.2u_1)$$

$$dVOC/dt = 0.05(100 - VOC) + 0.10 \max(T - 30, 0)$$

Fertiligation delivers ions via pump (u_1); crop transpires water/nutrients; drainage D leaches excess salts.

6. LINEARIZATION & VALIDATION

A (11×11), B (11×4)

Central Difference Jacobians ($h = 10^{-6}$):

$$A(:, i) = [f(x_0 + h e_i) - f(x_0 - h e_i)] / (2h)$$

$$B(:, j) = [f(u_0 + h e_j) - f(u_0)] / h \quad (u \in [0, 1])$$

Step Perturbation Verification:

$$\text{Step } \delta u_1 = +10\% \text{ pump over 4h:}$$

$$\delta M_L = +3.398\%, \quad \delta M_{NL} = +3.398\%$$

$$\|\delta x_{NL} - \delta x_L\|_{\infty} < 10^{-9}\% \text{ (Exact local consistency!)}$$

LINEARIZATION IS LOCAL — NOT GLOBAL:
A and B change with operating point, soil substrate, and growth phase!

8. DUAL OBSERVABILITY PROOF

4 vs 11 Transducers

Case A: 4 Primary Air Sensors:

$$C_{\text{prim}} = [I_{4 \times 4}, 0_{4 \times 7}] \Rightarrow \text{rank}(Ob) = 4 / 11$$

Nutrients/Tank Unobservable

Air sensors provide zero mathematical observability of root NPK or water storage.

Case B: Dedicated 11-Transducer Suite:

$$C_{\text{full}} = I_{11 \times 11} \Rightarrow \text{rank}(Ob) = 11 / 11$$

100% Fully Observable

Examiner Proof: Mathematically proves dedicated agrochemical probes are mandatory for greenhouse nutrient control!

10. SAFETY & 24H BENCHMARK

Interlocks & Scorecard

Safety Interlocks & Sensor Noise:

• Sensed: $y_{\text{meas}} = \text{clamp}(x + v, \min, \max)$, $v \sim N(0, \sigma^2)$

• Actuator Saturation: $u_{\text{act}} = \text{clamp}(u_{\text{cmd}}, 0, 1)$

• Emergency Interlock: IF $W < 10\%$ THEN $u_{\text{act}}(1) = 0$

Metric	PID	PP	Optimal LQR-I
Total ISE	1.53×10^8	8.04×10^8	1.57×10^7 (-89.7%)
Total IAE	50,307	118,518	5,854
TV (Chatter)	2,090.9	277.9	728.3
Mean CCI	99.76%	98.37%	99.94%

Crop Condition Index: $CCI = 0.50 \cdot CCI_{\text{env}} + 0.25 \cdot CCI_{\text{soil}} + 0.25 \cdot CCI_{\text{nutr}} = 99.94\%$

1. Plant & Sensors

11 States, ESP32 Layer

2. Botanical Stages

5 FAO-56 Phases

3. Nonlinear ODEs

Sandy, Loamy, Clay

4. Equilibrium Trim

$\|dx_0\| = 1.25e-10$

5. Linearization

Central Diff Jacobians

6. Modal Diagnostics

Ctrl 4/11 • Obsv 11/11

7. Tri-Hybrid Control

PID / PP / LQR-I CARE

8. Safety & Interlock

$W < 10\%$ Pump Cutoff

9. 24h Benchmark

CCI 99.94% • -89.7% ISE